

Project Report

Statistical Analysis of Rainfall Trends and Seasonal Variability in North Adelaide

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Abstract

The inter-annual variability of rainfall has a profound impact on urban drainage systems, flood management, water resource management, and environmental sustainability, making it an important hydrological factor. A statistical analysis of long-term rainfall and seasonal variability of rainfall in North Adelaide is performed in this study using monthly rainfall data obtained from the Bureau of Meteorology (BOM) Climate Data Online database that encompasses records of monthly rainfall from North Adelaide Station 023011 for the period 1996 to 2025. Descriptive statistical methods and graphical representations including bar charts, line graphs, box plots, and large-scale rainfall time series trend analysis were used to define rainfall patterns and identify hydrological trends. The analysis of quantitative data showed a large variability in annual rainfall through the duration of the study. The year with the highest total annual rainfall recorded over the duration was 2016 with approximately 737 mm, while the lowest total was 2006 with approximately 270.4 mm. Seasonal analysis found a large amount of the total annual rainfall, particularly during the months of June and July peak months for winter rainfall, whereas drought-stricken summer periods received comparatively less overall rainfall total. The distribution of rainfall data was positively skewed with infrequent occurrences of extreme rainfall. The results of this study confirm the characteristics of Adelaide's Mediterranean climate and illustrate the importance of analysing the variability of rainfall for the management of stormwater, flood mitigation and the design of climate resilient infrastructure.

Keywords

Rainfall variability, North Adelaide, Hydrological analysis, climate variability, statistical analysis, Bureau of Meteorology

GitHub Repository

[roshansapkota777/Rainfall-Analysis-North-Adelaide: Engineering Data Analytics project analysing rainfall trends and seasonal variability in North Adelaide](https://github.com/roshansapkota777/Rainfall-Analysis-North-Adelaide: Engineering Data Analytics project analysing rainfall trends and seasonal variability in North Adelaide)

1. Introduction

Variability in precipitation affects hydrological systems; urban drainage systems; environmental sustainability; and management of water resources. For example, in metropolitan areas, the amount and timing of rainfall affect how quickly stormwater runoff is produced, how often floods occur, how much groundwater is recharged, and how much infrastructure can withstand these events. Thus, having knowledge about patterns of precipitation is critical for engineers, environmental planners, and legislators.

Due to various climatic influences, including atmospheric circulation, Australia experiences considerable variability in its rainfall amounts. Gallant et al. (2007) note that the volume of rainfall in Australia can significantly differ in time and space due to climate oscillation and

seasonal weather systems. Furthermore, Westra et al. (2013) found an increase in the global frequency of extreme precipitation events, highlighting the need to analyze rainfall patterns as part of studies addressing climate and engineering.

Adelaide experiences a Mediterranean climate characterised by cool, wet winters and warm, dry summers. The seasonal variability of rainfall in Adelaide has a significant impact on urban drainage, stormwater management, and planning for water allocations. Urbanisation increases the importance of hydrological analysis because of the amount of runoff generated from impermeable surfaces and because of reduced infiltration due to urbanisation. In engineering, hydrological analysis is performed through studying rainfall trends and performing statistical evaluations to understand rainfall patterns, analyse variability in climate, and assist with the development of sustainable infrastructure. Statistically analysing rainfall data also assists engineers in identifying instances of extreme rainfall that could contribute to increased flood risk and increased load on our existing infrastructure. This research will conduct a statistical analysis of annual and monthly rainfall variability in North Adelaide using Bureau of Meteorology rainfall data collected between 1996 and 2025. The objective of the study is to identify seasonal rainfall patterns, long-term trends in rainfall amounts, variability between individual years, and those years with extreme amounts of rainfall and those with very little rainfall through descriptive statistics and graphical methods.

2. Data

2.1 Data Source

This study was based on data from the Bureau of Meteorology (BOM) Climate Data Online database. Monthly rainfall data were obtained from the North Adelaide weather station (023011) located in South Australia at latitude 34.92°S and longitude 138.6°E, and elevation 26 metres above sea level.

The data consisted of monthly rainfall data in mm from 1996 to 2025. A total of over 360 observations of monthly rainfall were analysed.

2.2 Study Area

North Adelaide is in Adelaide metropolitan region in South Australia and has a Mediterranean climatic regime. The region experiences cool, wet winters and dry summers, because of seasonal changes in atmospheric circulation systems. The stormwater systems within the region are impacted by rainfall variability, as is urban flooding, vegetation growth, and water resource management systems.

2.3 Data Cleaning and Preparation

Prior to analysis, data was cleaned to enhance data reliability and consistency. The cleaning process consisted of the following:

- Sampling measurements in the weather station grid.
- Knowledge of average annual rainfall.

- Development of new time series of precipitation
- Monthly and annual rainfall categories.
- Organisation of monthly and annual rainfall categories

The cleaned data set was segregated according to

- Monthly rainfall data
- Seasonal rainfall data,
- Annual total rainfall

This enhanced the efficiency of statistical interpretation and graphical analysis.

1. Methods

Rainfall behaviour and hydrological variability was assessed in North Adelaide using several statistical and graphical analytical techniques. The chosen methods were deemed to be appropriate, since the study was aimed at exploratory statistical analysis and climate trend analysis instead of predictive climate modelling.

3.1 Descriptive Statistical Analysis

The rainfall characteristics were summarised by descriptive statistical analysis for the entire study period. Several statistics, such as mean rainfall, minimum rainfall, maximum rainfall, annual variability, and averages of the season were computed.

Statistical Measure	Value
Mean Annual Rainfall	535.8 mm
Maximum Annual Rainfall	737.0 mm
Minimum Annual Rainfall	270.4 mm
Wettest Year	2016
Driest Year	2006

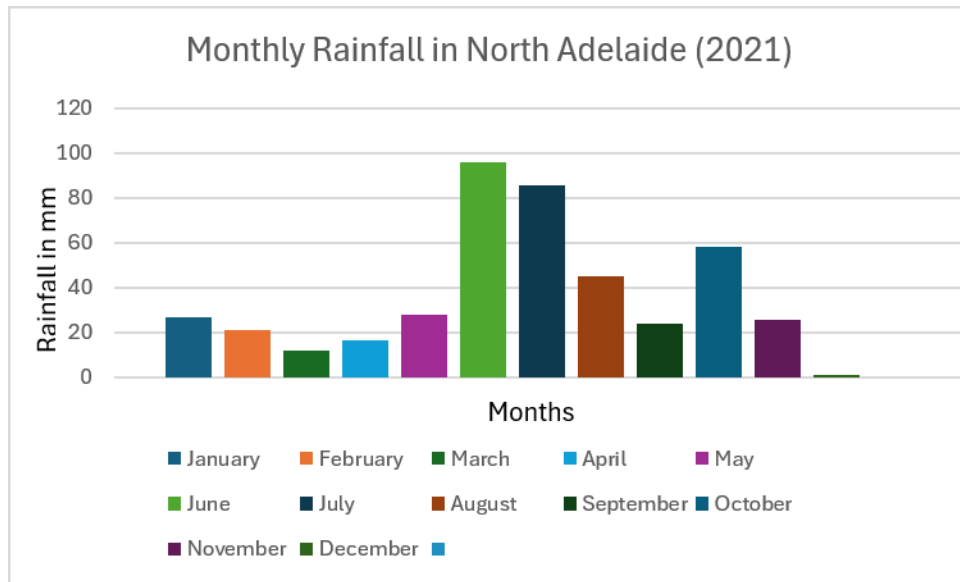
Descriptive statistics was chosen as they provide a quantitative summary of rainfall variability and intensity of precipitation.

3.2 Bar Chart Analysis

Rainfall intensity was compared to see the difference in the intensity of rain between each month and year using bar charts. This method was chosen because it illustrates the differences of rainfall magnitude and the distribution of rainfall in the season.

Figure 1. Monthly Rainfall distribution in 2021.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Rainfall(mm)	26.8	21.2	11.9	16.6	27.8	96.2	85.4	44.9	23.6	58.2	25.4	1.1



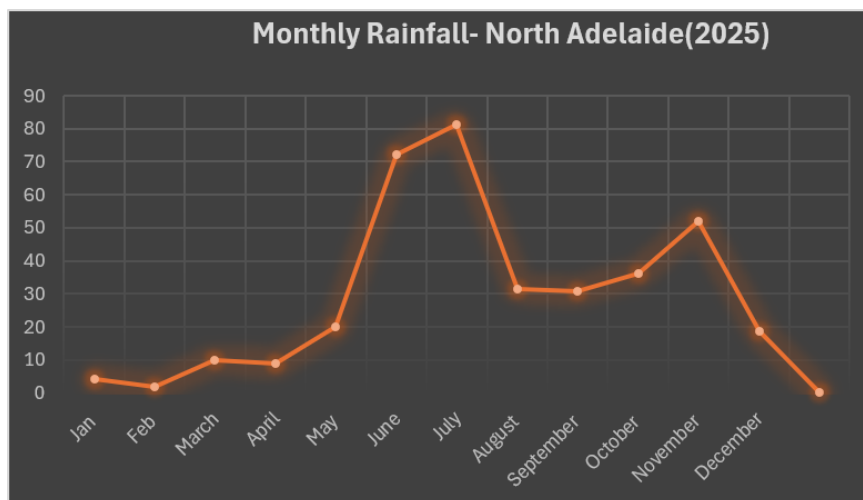
The above bar chart demonstrated significantly higher rainfall during winter months, particularly in June and July.

3.3 Line Graph Trend Analysis

Line graph analysis was carried out to find monthly rainfall variability and precipitation trends.

Figure 2 Monthly Rainfall trend in 2025

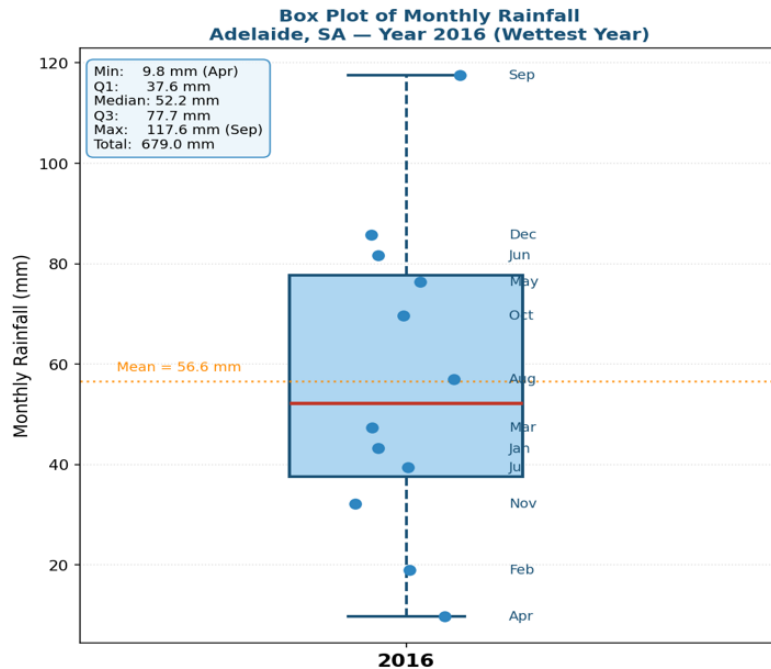
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Rainfall(mm)	4.3	1.8	9.8	8.8	20	72.3	81.3	31.6	30.7	36.3	51.9	18.5



The line graph analysis illustrated substantial inter-annual rainfall fluctuations and an increasing and decreasing rainfall trend.

3.4 Box- Plot Analysis

Box plot analysis was conducted to identify statistical characteristics like median, spread, variability, and potential outliers.



The box plot shows the spread of monthly rainfall across the 12 months of 2016. The median monthly rainfall is around 58 mm, meaning half of the months had rainfall below this value and half above it.

The box (Q1 to Q3) is relatively wide, indicating high variability between months. September was the wettest month with 117.6 mm, while April was the driest with only 9.8 mm.

Overall, the plot shows that 2016 had a wide range of monthly rainfall, with some months being very wet and others quite dry, which is typical of Adelaide's Mediterranean climate pattern.

3.5 Hypothesis framework

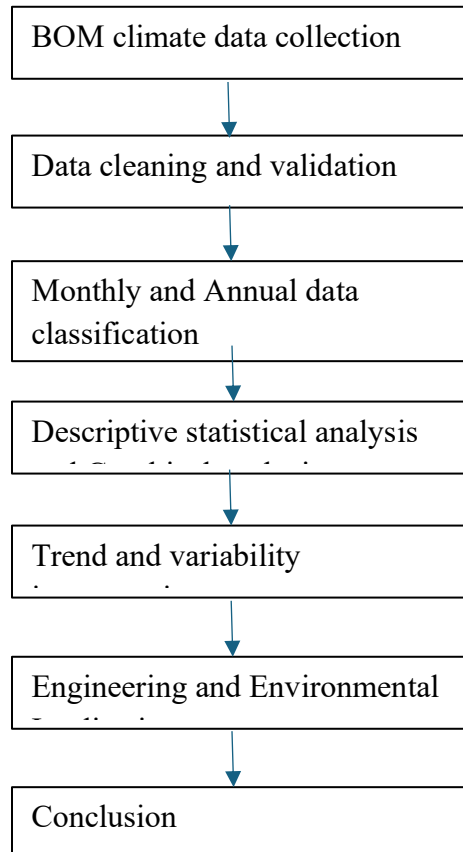
Null Hypothesis (H_0)

The distribution of rainfall over the month is not significantly different at different seasons in North Adelaide.

Alternative Hypothesis (H_1)

There is significant seasonal variation in monthly rainfall distribution within North Adelaide. The alternative hypothesis was supported by a graphical and descriptive statistical analysis, which revealed that the data presented a high level of variability across the season.

3.6 Methodology Flowchart



4. Results and Discussion

4.1 Annual Rainfall Variability

The annual rainfall analysis revealed significant inter-annual rainfall variability across North Adelaide in the 1996-2025 period. The rainfall varied significantly from one year to the next, which suggests an unstable rainfall pattern and hydrological conditions. The wettest year was 2016 and driest year was 2006. The average annual rainfall was approximately 535.8mm. The rainfall variability can introduce uncertainty in urban stormwater planning and design of flood mitigation infrastructure.

4.2 Seasonal Rainfall Behaviour

The rainfall was analyzed seasonally, and the result was a high level of dominance of rainfall in winter seasons throughout the study period. June and July saw the highest rainfall values for the whole year. The rainfall data analysis for the year 2021, shown in the bar chart, indicated that the amount of rainfall in the month of June was about 96.2 mm in the month of July was about 85.4mm respectively. Compared to this, December had only 1.1mm of rainfall, which was very dry for the summer.

In Adelaide, the Mediterranean weather characteristics are confirmed by these findings, as the winter frontal systems are a major source of annual precipitation.

4.3 Engineering and Hydrological Implications

The variability of rainfall directly impacts on hydrological behaviour and urban infrastructure performance in Adelaide. Hydrological events such as extreme rainfall can result in urban flooding, increase runoff generation, and overloading of drainage systems. The result of this research is relevant to stormwater drainage design, flood mitigation planning, sustainable urban development, climate resilience planning, water resource management and managing rainfall variability helps engineers and planners to make better adaptation plans for infrastructure, which consider climate variability.

5. Conclusion

The study statistically analysed the rainfall trends and seasonal variability for North Adelaide based on the Bureau of Meteorology rainfall data between 1996 and 2025.

Inter-annual rainfall variations were identified to be significant with high seasonal precipitation trends throughout the study period. The highest precipitation intensity was observed during the winter season especially during the June/July period and the summer season was comparatively dry. 2016 is the wettest year in the study period with around 737 mm of rainfall while 2006 is the driest year with around 270.4 mm of rainfall.

The graphical and statistical analysis showed that Adelaide had Mediterranean climate characteristics and that there was a significant inter-annual variability in rainfall. The results have implications and applications for hydrological and engineering studies that would benefit such projects as: stormwater management, flood mitigation infrastructure, climate adaptation planning, water quality assessments, and Allocation of water resources and water.

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[Climate Data Online - Map search](#)

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